

厦门大学《微积分 IV》课程试卷



_____学院_____系_____年级_____专业

学年学期: _____主考教师: _____ A 卷(√) B 卷()

一、求下列极限 (每小题 5 分, 共 30 分)

1. $\lim_{n \rightarrow \infty} \frac{n^{2025}}{(2n^{2024} + 1)(n + 2025)}$

解: 原式 $= \lim_{n \rightarrow \infty} \frac{1}{(2 + \frac{1}{n^{2024}})(1 + \frac{2025}{n})} = \frac{1}{2}$.

2. $\lim_{x \rightarrow 4} \frac{\sqrt{x} - 2}{x^2 + x - 20}$

解: 原式 $= \lim_{x \rightarrow 4} \frac{\sqrt{x} - 2}{(x - 4)(x + 5)}$
 $= \lim_{x \rightarrow 4} \frac{x - 4}{(x - 4)(x + 5)(\sqrt{x} + 2)} = \lim_{x \rightarrow 4} \frac{1}{(x + 5)(\sqrt{x} + 2)}$
 $= \frac{1}{36}$

3. $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\ln \sin x}{(\pi - 2x)^2}$

解: 原式 $= \lim_{x \rightarrow \frac{\pi}{2}} \frac{\frac{1}{\sin x} \cdot \cos x}{2(\pi - 2x) \cdot (-2)}$
 $= \lim_{x \rightarrow \frac{\pi}{2}} \frac{\cot x}{-4(\pi - 2x)} \stackrel{(\frac{0}{0})}{=} \lim_{x \rightarrow \frac{\pi}{2}} \frac{-\csc^2 x}{8} = -\frac{1}{8}$

4. $\lim_{x \rightarrow 0} (\frac{\sin x}{x^3} - \frac{1}{x^2})$.

解: 原式 $= \lim_{x \rightarrow 0} \frac{\sin x - x}{x^3} = \lim_{x \rightarrow 0} \frac{\cos x - 1}{3x^2} = \lim_{x \rightarrow 0} \frac{-\frac{1}{2}x^2}{3x^2} = -\frac{1}{6}$

$$5. \lim_{x \rightarrow e} (\ln x)^{\frac{1}{1-\ln x}}$$

$$\text{解: 原式} = \lim_{x \rightarrow e} (\ln x)^{\frac{1}{1-\ln x}} = \lim_{x \rightarrow e} (1 + \ln x - 1)^{\frac{-1}{\ln x - 1}} = e^{-1}$$

$$6. \lim_{x \rightarrow 0} \frac{\sqrt{1+x^3}-1}{\sqrt{1-\sin x}-\sqrt{1-\tan x}}$$

$$\text{解: 原式} = \lim_{x \rightarrow 0} \frac{\frac{1}{2}x^3}{\sqrt{1-\sin x}-\sqrt{1-\tan x}}$$

$$= \frac{1}{2} \lim_{x \rightarrow 0} \frac{x^3(\sqrt{1-\sin x} + \sqrt{1-\tan x})}{\tan x - \sin x}$$

$$= \frac{1}{2} \lim_{x \rightarrow 0} \frac{x^3(\sqrt{1-\sin x} + \sqrt{1-\tan x})}{\tan x(1-\cos x)} = \frac{1}{2} \lim_{x \rightarrow 0} \frac{x^3(\sqrt{1-\sin x} + \sqrt{1-\tan x})}{\frac{1}{2}x^3} = 2$$

二、计算下列积分：（每小题 5 分，共 35 分）

$$1. \int \left(\frac{1}{1+x^2} + x^{\frac{2}{3}} \right) dx$$

$$\text{解: 原式} = \arctan x + \frac{3}{5}x^{\frac{5}{3}} + c$$

$$2. \int \frac{1}{x^2-2x-3} dx$$

$$\begin{aligned} \text{解: 原式} &= \int \frac{1}{(x+1)(x-3)} dx = \frac{1}{4} \int \left(\frac{1}{x-3} - \frac{1}{x+1} \right) dx \\ &= \frac{1}{4} \ln(x-3) - \frac{1}{4} \ln(x+1) + c \end{aligned}$$

$$3. \int \sin^2 x \cdot \cos^3 x dx.$$

$$\text{解: 原式} = \int \sin^2 x \cdot \cos^2 x d \sin x$$

$$= \int \sin^2 x (1 - \sin^2 x) d \sin x = \frac{1}{3} \sin^3 x - \frac{1}{5} \sin^5 x + c$$

$$4. \int \frac{1}{x^2} \ln x dx.$$

$$\text{解: 原式} = \int \ln x d\left(-\frac{1}{x}\right) = -\frac{1}{x} \ln x + \int \frac{1}{x^2} dx$$

$$= -\frac{1}{x} \ln x - \frac{1}{x} + C$$

$$5. \int \frac{1}{1+\sqrt[3]{x-1}} dx.$$

$$\begin{aligned} \text{解: } \int \frac{1}{1+\sqrt[3]{x-1}} dx &\stackrel{\text{令 } t=\sqrt[3]{x-1}}{=} 3 \int \frac{t^2}{1+t} dt \\ &= 3 \int \left(t-1+\frac{1}{1+t}\right) dt = 3\left[\frac{1}{2}t^2 - t + \ln(1+t)\right] + C \\ &= \frac{3}{2}(x-1)^{\frac{2}{3}} - 3\sqrt[3]{x-1} + 3\ln(1+\sqrt[3]{x-1}) + C \end{aligned}$$

$$6. \int_{-2}^2 (2+x)\sqrt{4-x^2} dx.$$

$$\begin{aligned} \text{解: 原式} &= 2 \int_{-2}^2 \sqrt{4-x^2} dx + \int_{-2}^2 x\sqrt{4-x^2} dx \\ &= 4\pi - \frac{1}{3} \left(\sqrt{4-x^2}\right)^3 \Big|_{-2}^2 = 4\pi \quad (\text{利用面积和奇偶性}) \end{aligned}$$

$$7. \int_1^4 \frac{1}{x^2} \sqrt{1+\frac{1}{x}} dx.$$

$$\begin{aligned} \text{解: 原式} &= \int_1^4 -\sqrt{1+\frac{1}{x}} d\left(1+\frac{1}{x}\right) \\ &= -\frac{2}{3} \left(1+\frac{1}{x}\right)^{\frac{3}{2}} \Big|_1^4 = -\frac{2}{3} \left[\left(\frac{5}{4}\right)^{\frac{3}{2}} - 2^{\frac{3}{2}}\right] \end{aligned}$$

三、求函数的导数或微分：（每小题 5 分,共 20 分）

1. $y = \ln(1+x) + \tan x + \sin \frac{\pi}{3}$, 求 y' .

解: $y' = \frac{1}{1+x} + \sec^2 x + 0$

2. $y = \sqrt[3]{2x + e^{3x}}$, 求 dy .

解: $y' = \frac{1}{3}(2x + e^{3x})^{-\frac{2}{3}}(2 + 3e^{3x})$

$$dy = \frac{1}{3}(2x + e^{3x})^{-\frac{2}{3}}(2 + 3e^{3x})dx$$

3. 设 $y = (\cos x)^{\sin x}$, 求 $\frac{dy}{dx}$.

解: 两边同时取对数: $\ln y = \sin x \ln \cos x$

两边同时对 x 求导得:

$$\frac{1}{y} \frac{dy}{dx} = \cos x \ln \cos x + \sin x \frac{1}{\cos x} (-\sin x)$$

整理得: $\frac{dy}{dx} = (\cos x)^{\sin x} (\cos x \ln \cos x - \tan x \cdot \sin x)$

4. 设 $f(x)$ 为奇函数且有 $f'(-x_0) = -3$, 问 $f'(x_0)$ 是否一定存在, 若存在请求出.

解: $f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$

$$\stackrel{\text{令 } t = -x}{\Rightarrow} = \lim_{t \rightarrow -x_0} \frac{f(-t) - f(x_0)}{-t - x_0} = \lim_{t \rightarrow -x_0} \frac{-f(t) + f(-x_0)}{-t - x_0}$$

$$= \lim_{t \rightarrow -x_0} \frac{f(t) - f(-x_0)}{t - (-x_0)} = f'(-x_0) = -3$$

四、(7分) 证明: 当 $0 < x < \frac{\pi}{2}$ 时, 有 $\sin x + \tan x > 2x$.

证明：设 $f(x) = \sin x + \tan x - 2x$,

$$\begin{aligned} \text{则 } f'(x) &= \cos x + \frac{1}{\cos^2 x} - 2 = \frac{\cos^3 x + 1 - 2\cos^2 x}{\cos^2 x} = \frac{[\cos^2 x(\cos x - 1) + (1 - \cos x)(1 + \cos x)]}{\cos^2 x} \\ &= \frac{(1 - \cos x)(1 + \cos x - \cos^2 x)}{\cos^2 x} \end{aligned}$$

当 $0 < x < \frac{\pi}{2}$ 时, $0 < \cos x < 1, (1 + \cos x - \cos^2 x) > 0$, 故 $f'(x) > 0$, 从而 $f(x)$ 严格单增, 故

$f(x) > f(0) = 0$, 即 $\sin x + \tan x > 2x$.

(另解: 当 $0 < x < \frac{\pi}{2}$ 时,

$$f'(x) = \cos x + \frac{1}{\cos^2 x} - 2 > \cos^2 x + \frac{1}{\cos^2 x} - 2 = \left(\cos x - \frac{1}{\cos x}\right)^2 \geq 0.)$$

五、(8 分) 求函数 $y = \frac{x^3}{2(x-1)^2}$ 的单调区间、极值点.

$$\text{解: } y' = \frac{x^2(x-3)}{2(x-1)^3},$$

在 $x_1 = 1$ 处, 函数没有定义, 令 $y' = 0$ 得驻点 $x_2 = 0$, $x_3 = 3$.

x	$(-\infty, 0)$	0	$(0, 1)$	1	$(1, 3)$	3	$(3, +\infty)$
$f'(x)$	+	0	+	没定义	-	0	+
$f(x)$	↗	非极值 0	↗	没定义	↘	极小值 27/8	↗

从表中可以看到, 1) 函数 $y = \frac{x^3}{2(x-1)^2}$ 在 $(-\infty, 1)$ 和 $(3, +\infty)$ 单增, 在 $(1, 3)$ 上单减;

2) 在 $x = 3$ 处取得极小值, 极小值为 $\frac{27}{8}$